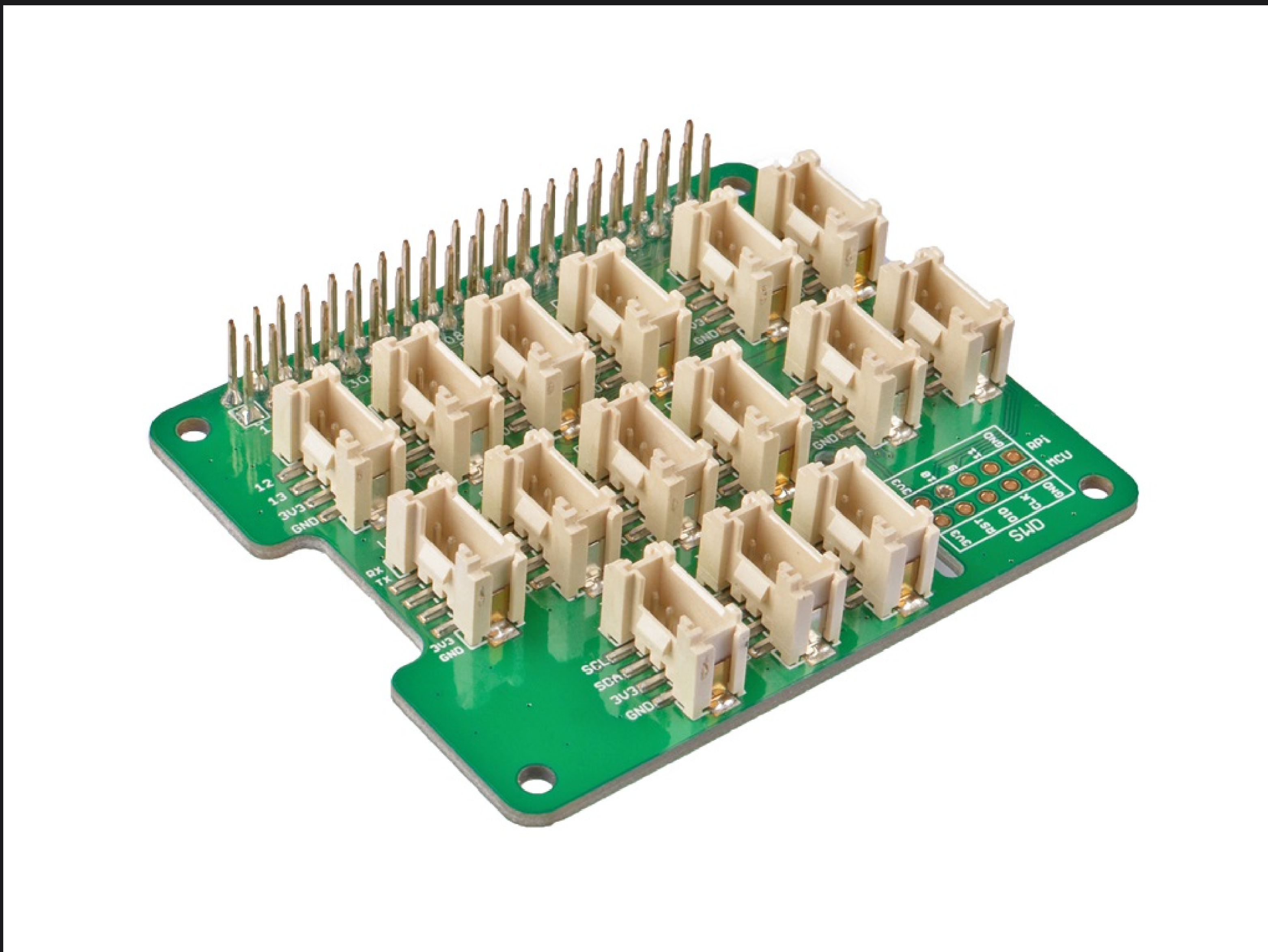
Grove Base Hat for Raspberry Pi



Today, the grove series of sensors, actuators, and displays have grown into a large family. More and more grove modules will join the whole Grove ecosystem in the future. We see the Grove helps makers, engineers, teachers, students and even artists to build, to make, to create...We always feel it is our responsibility to make the Grove module compatible with more platforms. Now we bring you the Grove Base Hat for Raspberry Pi and Grove Base Hat for Ras pberry Pi Zero, in another word, we bring the Raspberry Pi the whole Grove System.

The Grove Base Hat for Raspberry Pi provide Digital/Analog/I2C/PWM/UART port to meet all your needs. With the help of build-in MCU, a 12-bit 8 channel ADC is also available for Raspberry Pi.

Frankly speaking, it's about 60 Grove modules support the Grove Base Hat for Raspberry Pi now. However, we will continue to add new compatible modules, the more you use, the more grove added.

Get One Now 🖱️

Features

- Support Raspberry Pi 2/3B/3B+/4/Zero
- build-in MCU
- 12-bit ADC
- Multi-type Grove port

Specification

Item	Value
Operating Voltage	3.3V
MCU	STM32/MM32
ADC	12-bit 8 channel
Grove Port	6 Digital 4 Analog 3 I2C 1 PWM 1UART
Raspberry pi communication bus	I2C
I2C Address	0x04/0x08

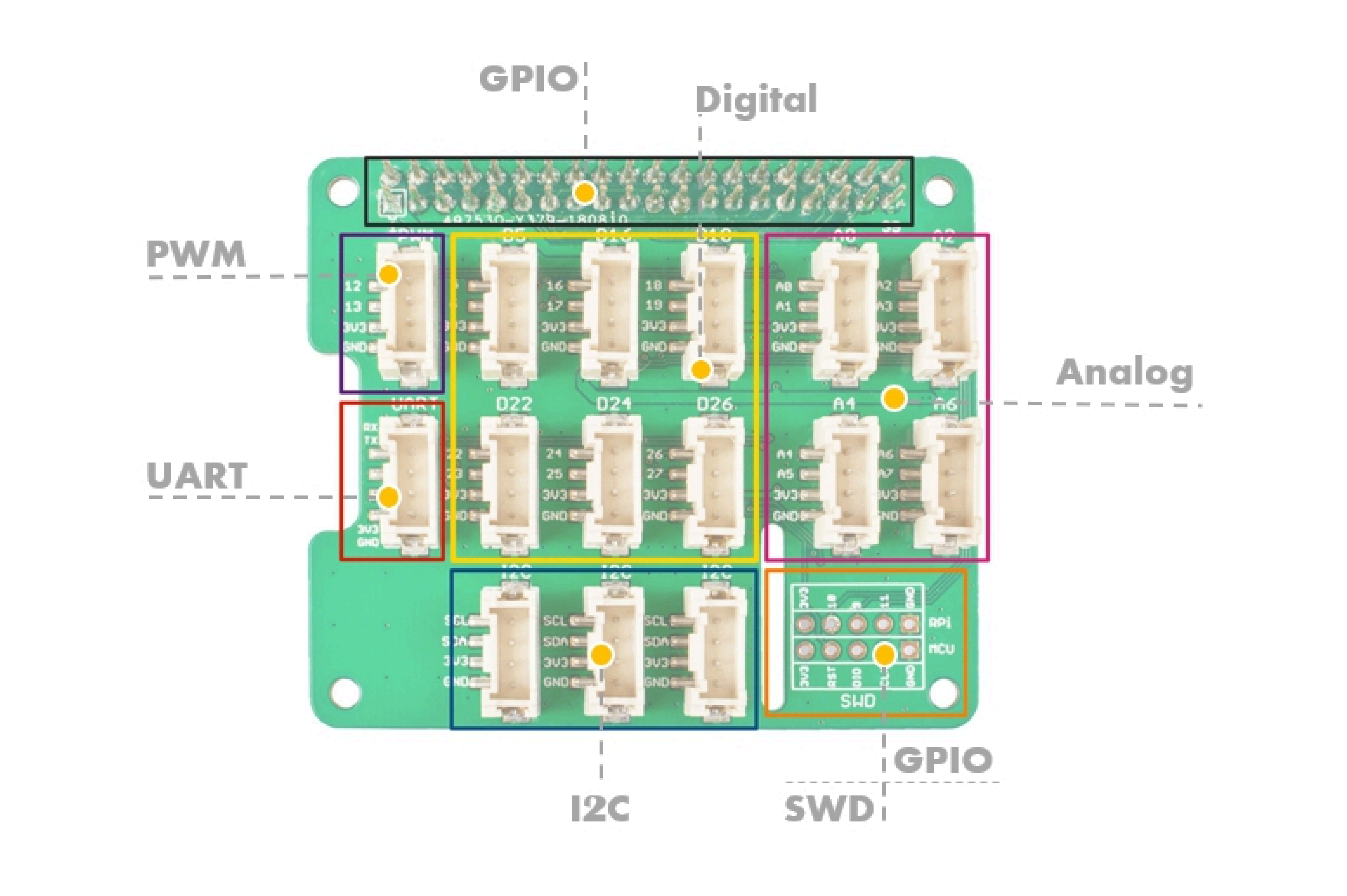
⚠ **CAUTION**

The operating voltage is 3.3V, please do not input more than 3.3V, otherwise it may damage the Raspberry Pi. Moreover, this hat can not work with 5V grove module via grove port, please use 3.3V compatible Grove module.

Hardware Overview

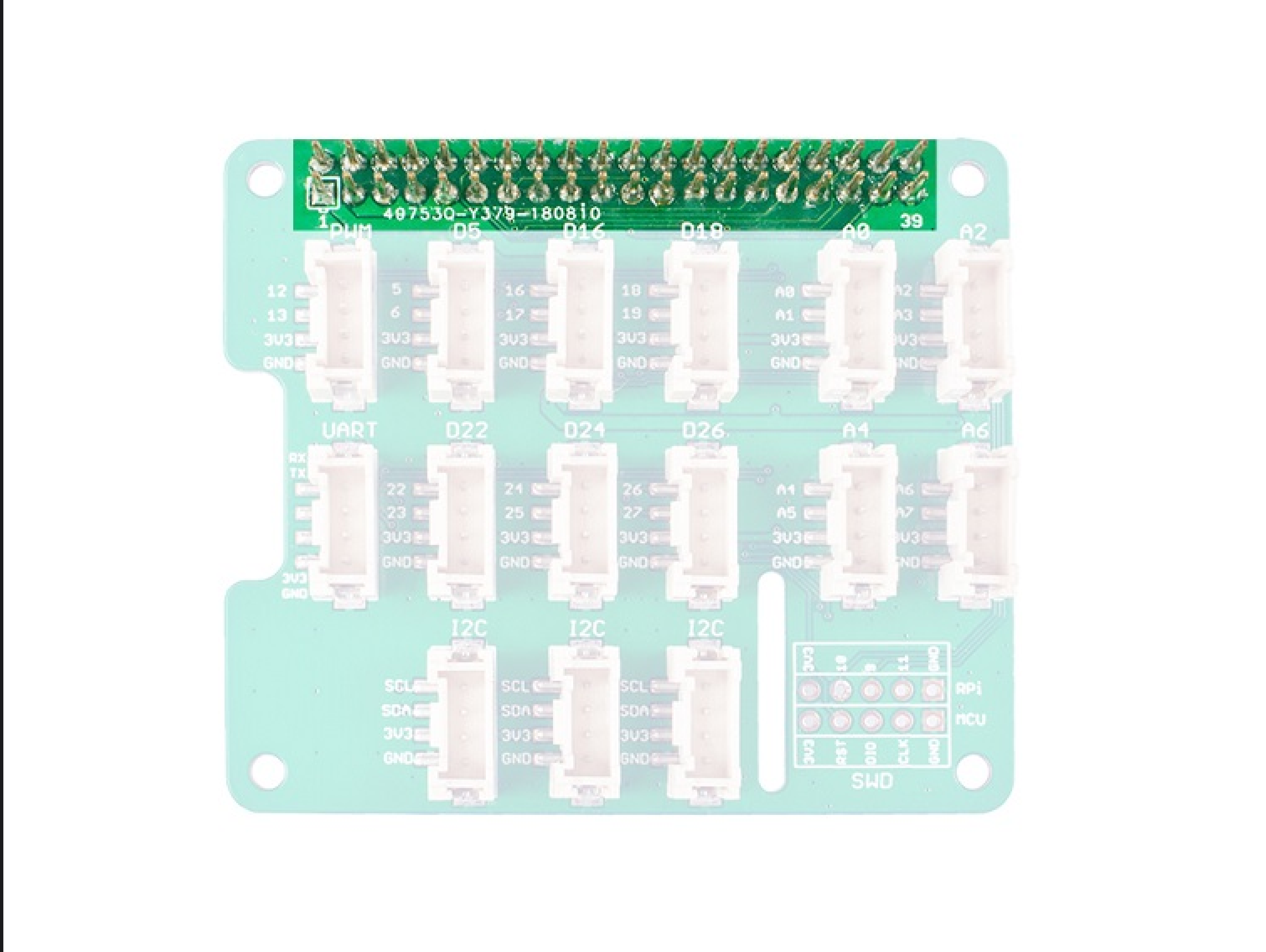
Pin Out

Overview



GPIO

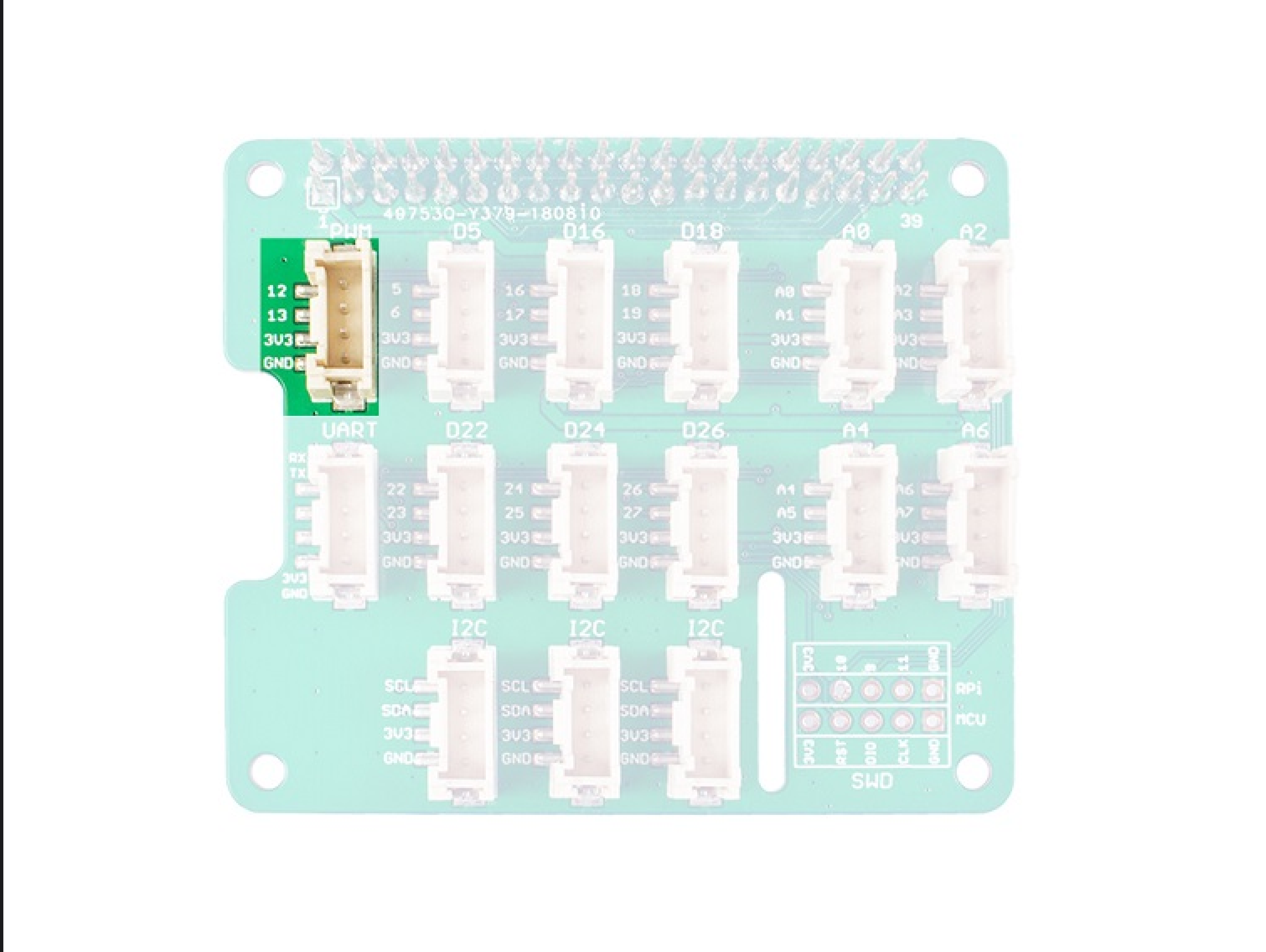
The same pin out as the raspberry pi.



PWM(pulse-width modulation)

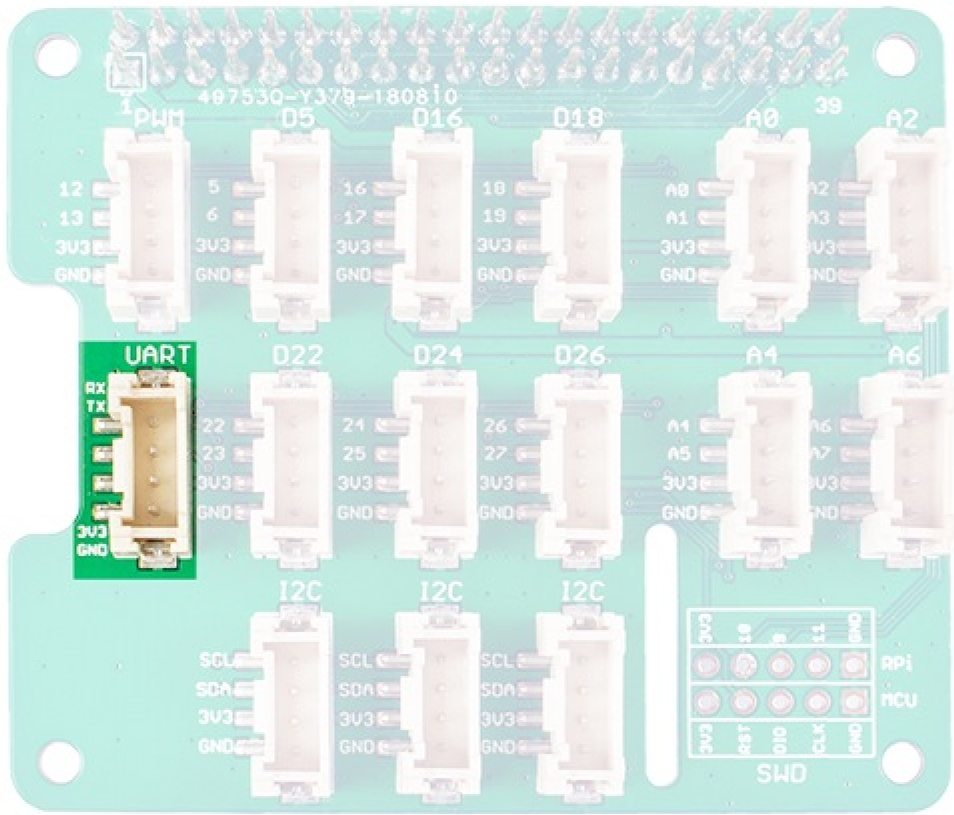
The Grove PWM Port connect to GPIO/BCM pin12(PWM0) and GPIO/BCM pin13(PWM1), which is the hardware PWM pin of Raspberry Pi, in addition, you can use all the GPIO pin as the soft PWM pin.

> **NOTE**
>
> 0- All the silkscreen layer pin number besides the Grove port is the BCM pin number. The difference between BCM pins and the physical pins please refer to here
>
> 1- Compared with hardware PWM, the software PWM isn't so accurate and will have trouble at high frequencies.
>
> 2- The GPIO/BCM pin18 is also marked as PWM0, actually the GPIO/BCM 12 and the GPIO/BCM 18 share the same PWM channel, so they can't set to different rate.
>
> 3- The audio jack output also uses PWM 0 and PWM 1, so you can't have audio output on that socket and use the PWMs at the same time.



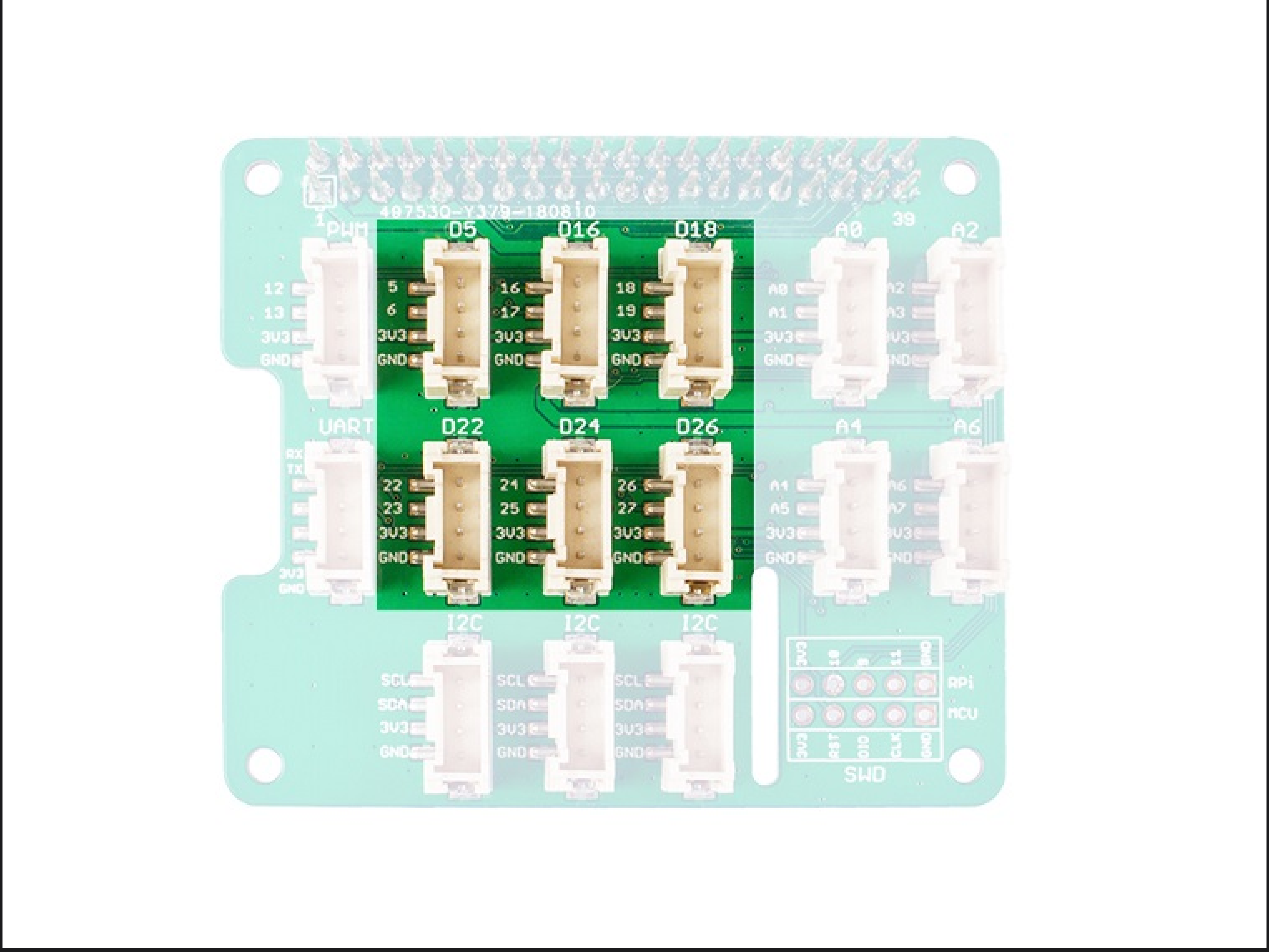
UART

The Grove UART port connect to the GPIO14(UART0 TX) and GPIO15(UART0 RX). UART is commonly used on the Pi as a convenient way to control it over the GPIO, or access the kernel boot messages from the serial console (enabled by default).It can also be used as a way to interface an Arduino, bootloaded ATmega, ESP8266, etc with your Pi.



Digital

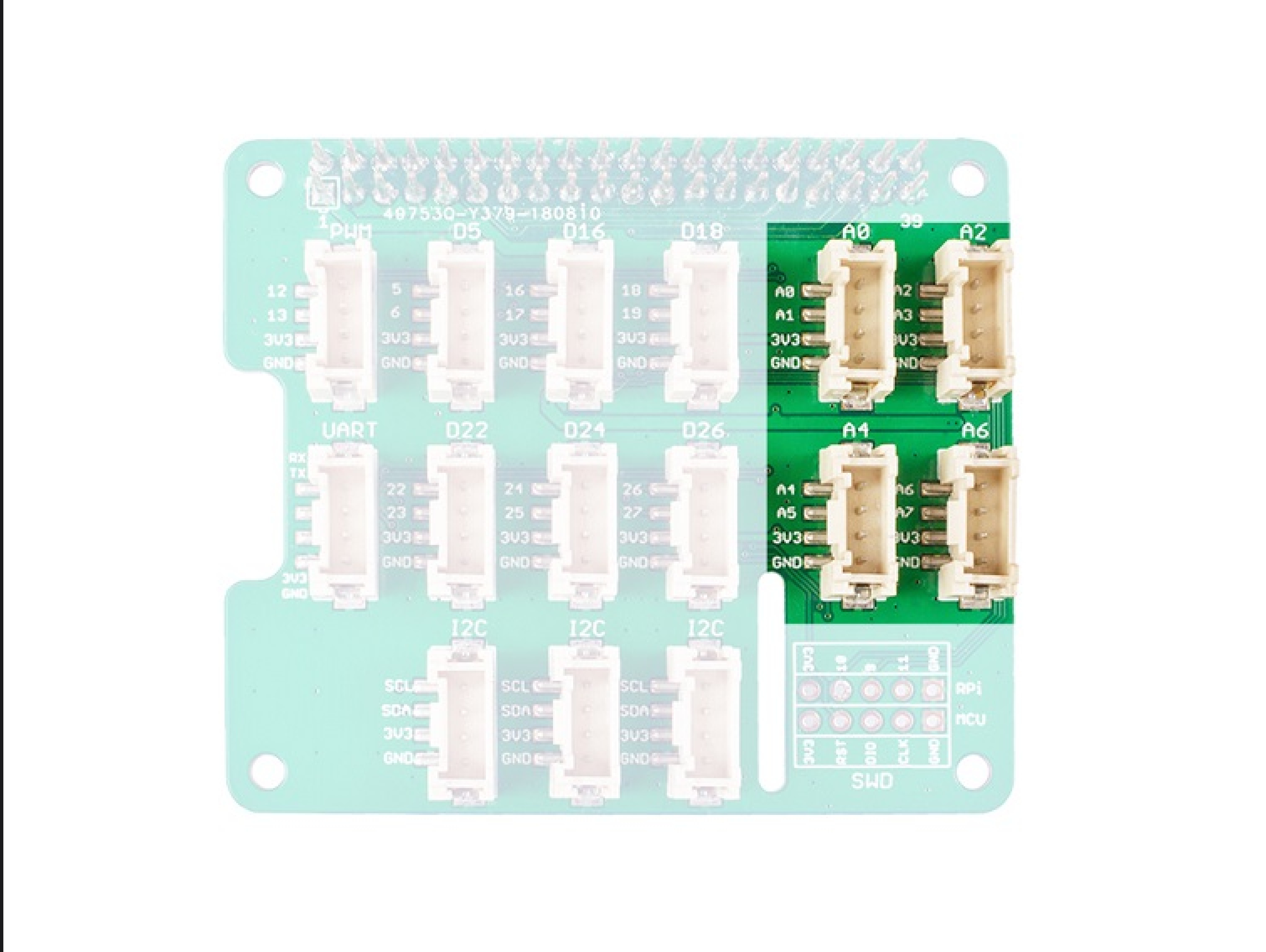
There are 6 digital Grove sockets in this board, normally the yellow wire(which connect to the top pin of the 4 pins Grove socket as) of Grove cable is the signal wire, so we name the digital Grove port **D5/D16/D18/D22/D24/D26**.



Analog

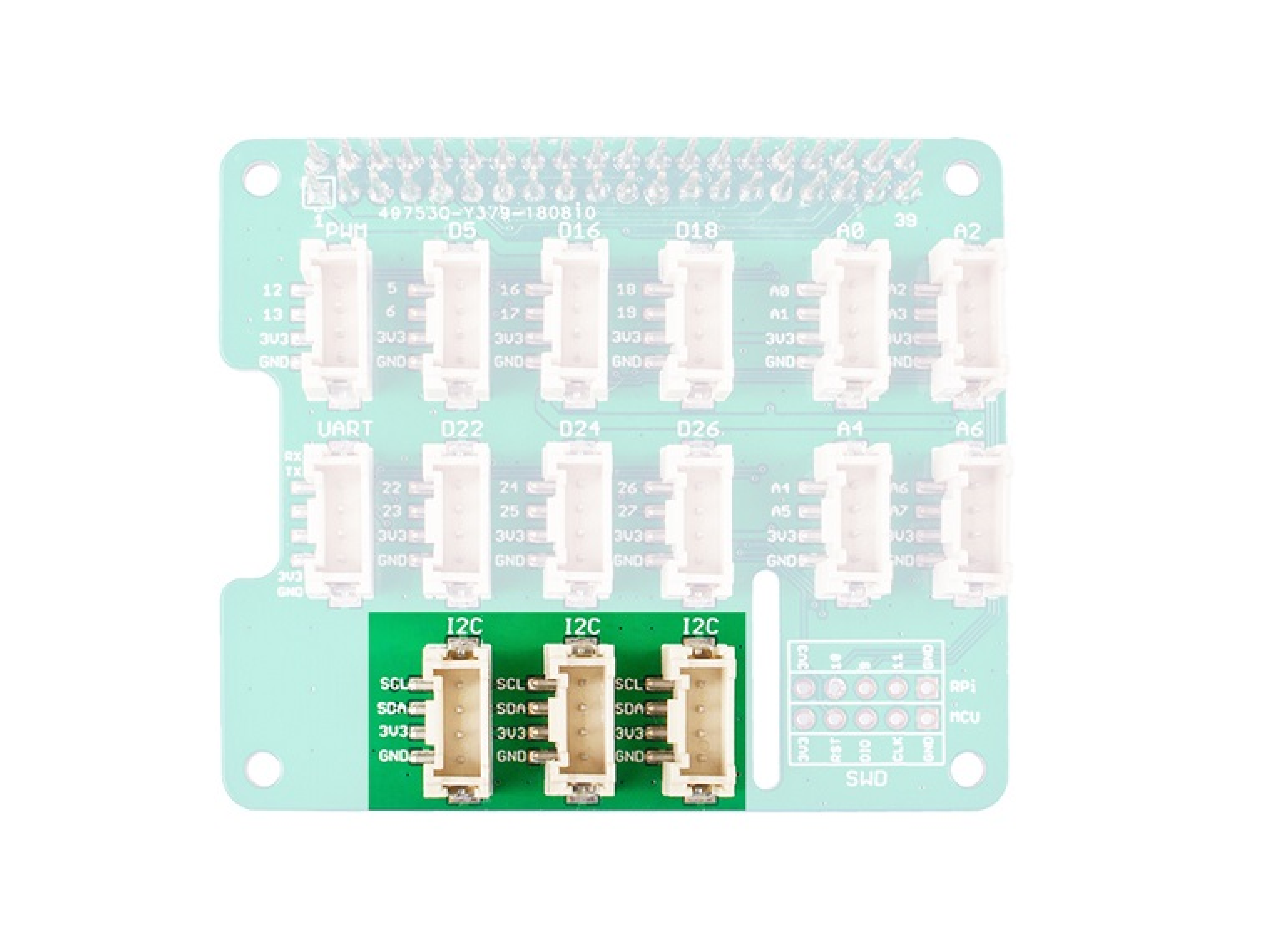
As we know, there is no ADC in the Raspberry Pi, so it can not work with analog sensor directly. Now with the help of the build-in MCU STM32, the Grove base hat can work as an external 12-bit ADC, which means you can use analog sensor with your Raspberry Pi. Even more pleasing is that not one but four analog Grove sockets are available.

The analog sensor inputs the analog voltage into the 12-bit ADC. After the ADC convert the analog data to digital data, it input the digital data to the Raspberry Pi through the I2C interface.



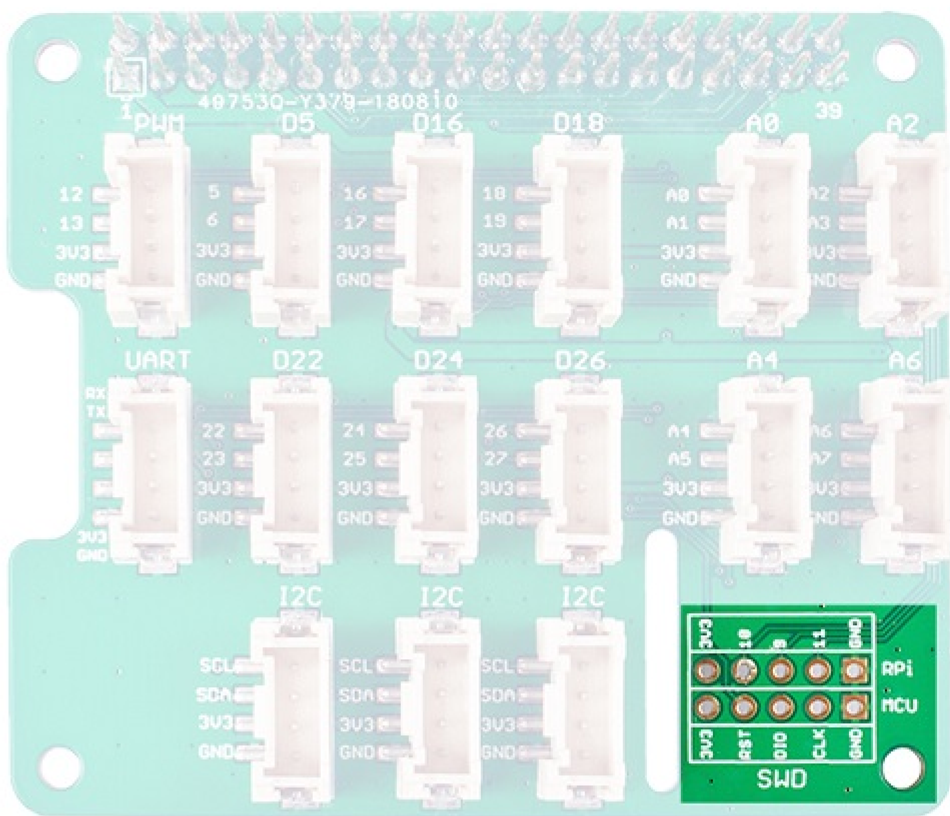
I2C

There are three I2C port available in this board, they all connect to the I2C pin of the raspberry directly. You can consider this part as an I2C hub. Most of seeed's new grove modules have I2C interface, you may find those three port is extremely useful.



SWD

We use SWD port to burn the firmware to this hat. In addition, you can see 3 GPIO pins in this section, i.e., **pin 9/pin 10/pin 11**. Those three pins do not used by any Grove port, you are free to use them without worrying about pin conflicts.



Grove Base Hat for Raspberry Pi Vs. GrovePi+

Parameter	Grove Base Hat for Raspberry Pi	GrovePi+
Working Voltage	3.3V	5V
MCU	STM32F030F4P6	ATMEGA328P
Grove Ports	6 x Digital(3.3V); 4 x Analog(3.3V) ; 3 x I2C(3.3V); 1 x PWM(3.3V) ; 1 x RPISER(UART) connect to Raspberry Pi(3.3V) ; 1 x SWD	7 x Digital(5V); 3 x Analog(5V) ;3 x I2C(5V) ;1 x SERIAL: Connect to ATMEGA328P D0/1(5V) ; 1 x RPISER: Connect to Raspberry Pi(3.3V) ;1 x ISP

Parameter	Grove Base Hat for Raspberry Pi	GrovePi+
Grove-Digital	Connect to Raspberry Pi directly	Connect to ATMEGA328P digital pins and transfer to I2C signal, then through level converter to Raspberry Pi
Grove-Analog	Connect to STM32F030F4P6(12bit ADC) and then transfer to I2C signal,route to Raspberry Pi directly	Connect to ATMEGA328P analog pins(10bit ADC) and then transfer to I2C signal, then through level converter to Raspberry Pi
Grove-I2C	Connect to Raspberry Pi directly	Connect through level converter to Raspberry Pi
Grove-PWM	Connect to Raspberry Pi directly	NA
RPISER	Connect to Raspberry Pi directly	Connect to Raspberry Pi directly
SERIAL	NA	Connect to ATMEGA328P digital pins D0/D1 and transfer to I2C signal, then through level converter to Raspberry Pi
SWD	Burn firmware to STM32F030F4P6	NA
ISP	NA	Burn firmware to ATMEGA328P
Raspberry Pi Connector Pins	40	26

Getting Started

Hardware

Materials required

-- Raspberry Pi x1

-- Grove Base Hat for Raspberry Pi x1

-- Grove module

- **Step 1.** Plug the Grove Base Hat for Raspberry Pi into the Raspberry Pi.
- **Step 2.** Plug the Grove module into the corresponding Grove port.
- **Step 3.** Power the Raspberry Pi with micro-usb cable.

> **CAUTION**
>
> We recommend that you power up the Raspberry Pis after all the hardware connections are complete. Please do not hot plug the sensor module, otherwise the Raspberry Pi may crash.

Software

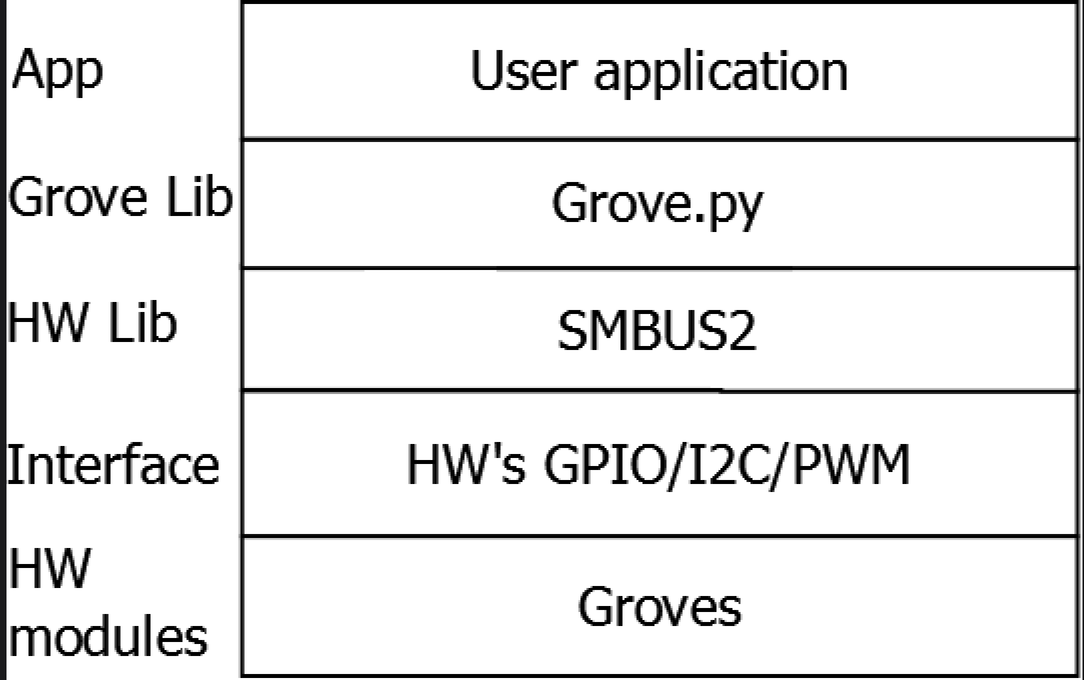
> **CAUTION**
>
> Currently, this board is available in two versions on the market: the STM32 version (V1.1) and the MM32 version (V1.0). The current shipping version is the STM32 version.
>
> If you experience issues using the Grove Base Hat for Pi with the `grove.py` library, please check your board version:
>
> 1. **If it is the STM32 version**, the I2C address is `0x04`. You need to change the I2C address in `/home/username/.local/lib/python3.9/site-packages/adc.py` to `0x04`.
> 2. **If it is the MM32 version**, the I2C address is `0x08`. You need to change the I2C address in `/home/username/.local/lib/python3.9/site-packages/adc.py` to `0x08`.

In this section we will introduce how to install the **seeed grove.py** library and how to use I2C, PWM, Digital and analog port of the Grove Base Hat for Raspberry Pi.

> **TIP**
>
> If you do not know how to use a raspberry pi, please check here before start.

Archtecture

To operate grove sensors, the grove.py depends many hardware interface libraries such as mraa/smbus2/rpi.gpi/rpi_ws281x.



Installation

> **TIP**
>
> A virtual environment is currently the most stable and recommended way to replicate in Bookworm.

Install Dependencies

Add repository

```
echo "deb https://seeed-studio.github.io/pi_repo/ stretch main" | sudo tee
/etc/apt/sources.list.d/seeed.list
```

Add public GPG key

```
curl https://seeed-studio.github.io/pi_repo/public.key | sudo tee
/etc/apt/trusted.gpg.d/seeed.gpg > /dev/null
sudo apt update
```

Install underlying dependencies (global is fine)

```
# Optional: Seeed binary package (not needed for most sensors)
sudo apt install libbmi088 libbma456
# Required: Python layer dependencies
pip install smbus2 pyserial seeed-python-dht   # DHT DHT series required
```

Enable I2C interface

```
sudo raspi-config
```

- Select interfacingg Options>I2C>Yes>Ok>Finish
- Enable I2C interface

Install required packages

> **TIP**
>
> In the lastest version of Python3, it is recommended to use *virtualenv* for isolated package management.

> **TIP**
>
> If you are using **Raspberry Pi with Raspberrypi OS >= Bullseye**, you have to use this command line **only with Python3**. The following instruction is working on Bookworm OS.

vertual environment

```
# Create once
mkdir ~/grove_env && cd ~/grove_env
python3 -m venv --system-site-packages env
# From now on, you need to run this first every time you open the terminal.
source ~/grove_env/env/bin/activate
```

Install grove.py

For beginner or library user only, please install with online method.

- Automatic dependency installation and library deployment.
- Quick to get started, no manual operation required.

For developer or advanced user, please install dependencies and then install grove.py with source code.

- Includes the complete `grove.py` repository, source code, examples, and documentation
- Suitable for viewing source code and examples, modifying library functions

For beginner or library user | For developer or advanced user

To install into the global environment, you can type the following command:

```
curl -sL https://github.com/Seeed-Studio/grove.py/raw/master/install.sh | sudo bash -s -
```

If you want to install into a virtual environment, first active your virtualenv and type the following command:

```
curl -sL https://github.com/Seeed-Studio/grove.py/raw/master/install.sh | bash -s -- --
user-local --bypass-gui-installation
```

> **NOTE**
> Due to updates to the Raspberry Pi, the one-click installation script is relatively old and may not work for one-click installation. It is recommended to use the second method - the more stable source code method to download the Grove library.

General Template for Running Routines (Every Time a New Terminal Opens)

Virtual environment

```
source ~/grove_env/env/bin/activate
cd ~/grove_env/grove.py/grove
python grove_xxx.py          # xxx = Module Name
```

Global environment

```
 grove_xxx.py          # xxx = Module Name
```

Usage

Now you can use the Grove Base Hat for Raspberry Pi with dozens Grove modules, tap the command **grove_** and press the ++tab++ key to check the supported Grove list.

```
(env) pi@raspberrypi:~ $ grove_
grove_12_key_cap_i2c_touch_mpr121
grove_16x2_lcd
```

```
grove_1wire_thermocouple_amplifier_max31850
grove_3_axis_accelerometer_adxl372
grove_3_axis_compass_bmm150
grove_3_axis_digital_accelerometer
grove_4_digit_display
grove_6_axis_accel_gyro_bmi088
grove_air_quality_sensor_v1_3
grove_button
grove_cap_touch_slider_cy8c
grove_collision_sensor
grove_current_sensor
grove_gesture_sensor
grove_gpio
grove_high_accuracy_temperature
grove_i2c_color_sensor_v2
grove_i2c_motor_driver
grove_i2c_thermocouple_amplifier_mcp9600
grove_imu_9dof_icm20600_ak09918
grove_lcd_1.2inches
grove_led
grove_light_sensor_v1_2
(env) pi@raspberrypi:~ $ grove_
grove_12_key_cap_i2c_touch_mpr121
grove_16x2_lcd
grove_1wire_thermocouple_amplifier_max31850
grove_3_axis_accelerometer_adxl372
grove_3_axis_compass_bmm150
grove_3_axis_digital_accelerometer
grove_4_digit_display
grove_6_axis_accel_gyro_bmi088
grove_air_quality_sensor_v1_3
grove_button
grove_cap_touch_slider_cy8c
grove_collision_sensor
grove_current_sensor
grove_gesture_sensor
grove_gpio
grove_high_accuracy_temperature
grove_i2c_color_sensor_v2
grove_i2c_motor_driver
grove_i2c_thermocouple_amplifier_mcp9600
grove_imu_9dof_icm20600_ak09918
grove_lcd_1.2inches
grove_led
grove_light_sensor_v1_2
grove_loudness_sensor
grove_mech_keycap
```

```
grove_mini_pir_motion_sensor
grove_moisture_sensor
grove_multi_switch
grove_multi_switch_poll
grove_oled_display_128x64
grove_optical_rotary_encoder
grove_piezo_vibration_sensor
grove_pwm_buzzer
grove_recorder_v3_0
grove_relay
grove_rotary_angle_sensor
grove_round_force_sensor
grove_ryb_led_button
grove_servo
grove_slide_potentiometer
grove_sound_sensor
grove_step_counter_bma456
grove_switch
grove_temperature_humidity_bme680
grove_temperature_humidity_sht31
grove_temperature_sensor
grove_thumb_joystick
grove_tilt_switch
grove_time_of_flight_distance
grove_touch_sensor
grove_ultrasonic_ranger
grove_uv_sensor
grove_water_sensor
grove_ws2813_rgb_led_strip
```

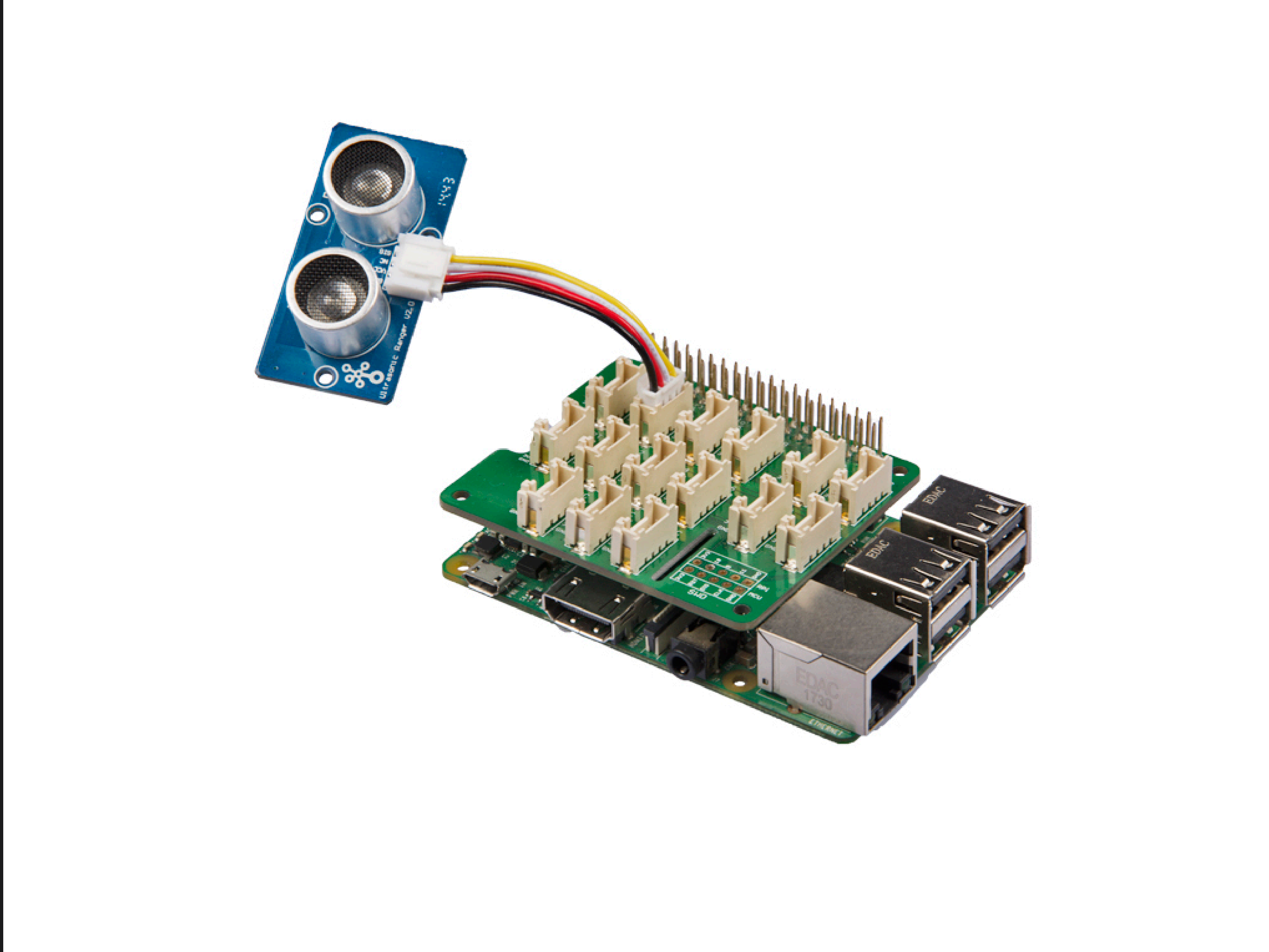
Then we will show you how to use them according to port type.

Digital Port

> **CAUTION**
>
> If you are using **Raspberry Pi with Raspberrypi OS >= Bullseye**, you have to use this command line **only with Python3**.

We will take the Grove - Ultrasonic Ranger for example to introduce the Digital port.

Hardware connection diagram



Tap the following command `grove_ultrasonic_ranger 5 6` in the command line interface.

```
pi@raspberrypi:~$  grove_ultrasonic_ranger 5 6
Detecting distance...
6.979909436456088 cm
7.966469074117726 cm
12.451204760321255 cm
15.184797089675378 cm
17.429220265355603 cm
18.73230112010035 cm
20.717752390894397 cm
19.83807004731277 cm
17.3059003106479 cm

^CTraceback (most recent call last):
  File "/usr/local/bin/grove_ultrasonic_ranger", line 11, in <module>
    load_entry_point('grove.py==0.5', 'console_scripts', 'grove_ultrasonic_ranger')()
  File "/usr/local/lib/python3.5/dist-packages/grove/grove_ultrasonic_ranger.py", line
107, in main
```

```
    time.sleep(1)
KeyboardInterrupt
```

When you change the distance between the Grove - Ultrasonic Ranger and the target object, the measurment value will change accordingly. Press ++ctrl+c++ to quit.

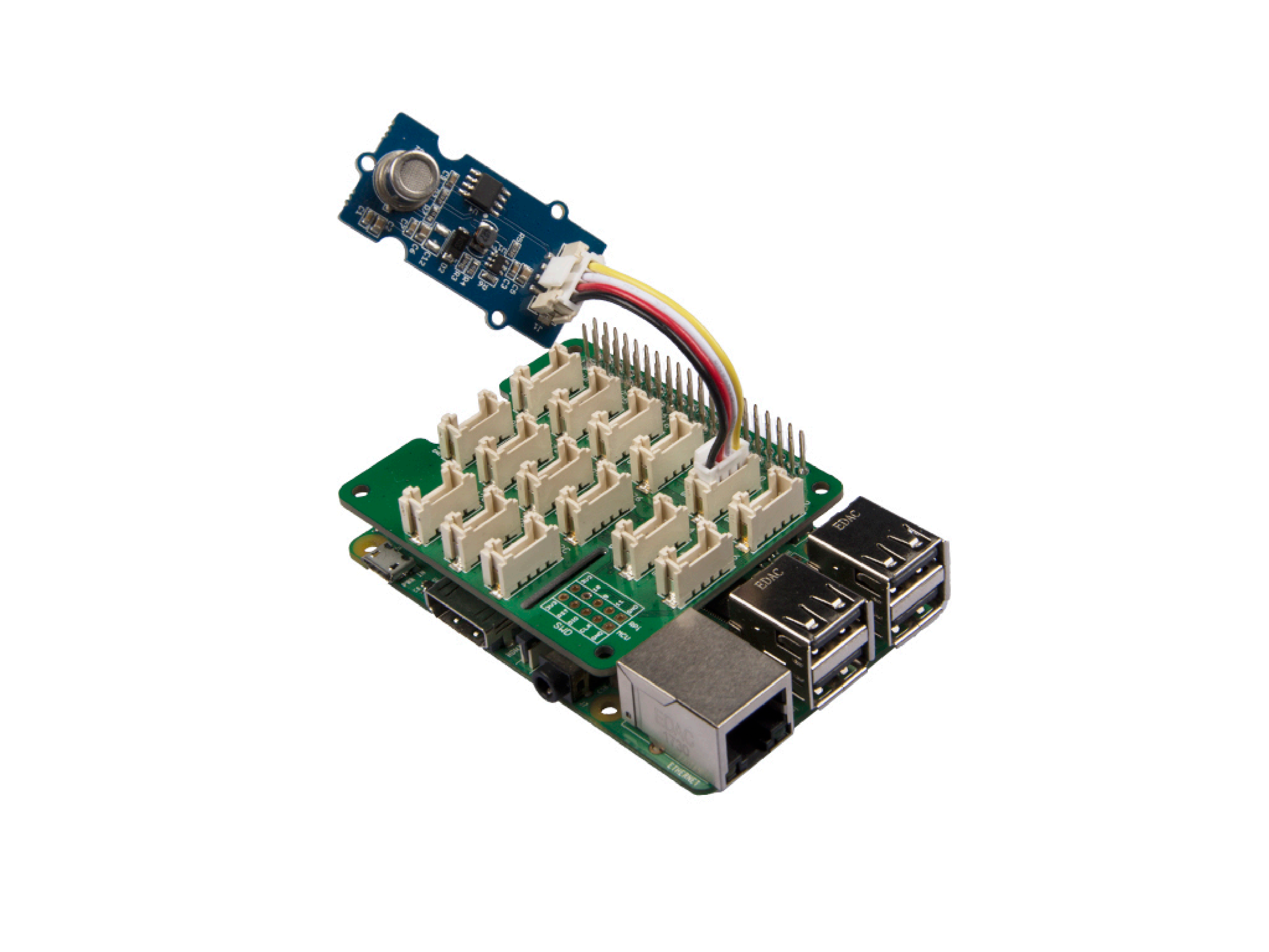
> **CAUTION**
>
> For most grove module, you need to add the pin number parameter, like `grove_ultrasonic_ranger 5 6`, **5** and **6** are the GPIO/BCM pin. However, you may have noticed that in the first example `grove_pwm_buzzer`, we didn't add a parameter after the command. This is because the PWM port and I2C port do not require pin number parameters. You can find the pin number silkscreen just besides the Grove sokect.

Analog Port

> **CAUTION**
>
> If you are using **Raspberry Pi with Raspberrypi OS >= Bullseye**, you have to use this command line **only with Python3**.

We will take the Grove - Air quality sensor v1.3 for example to introduce the Analog port.

Hardware connection diagram



Tap the following command `grove_air_quality_sensor_v1_3 0 1` in the command line interface.

```
pi@raspberrypi:~$  grove_air_quality_sensor_v1_3 0 1
Detecting ...
62, Air Quality OK.
63, Air Quality OK.
61, Air Quality OK.
61, Air Quality OK.
59, Air Quality OK.
62, Air Quality OK.
60, Air Quality OK.
60, Air Quality OK.
59, Air Quality OK.
60, Air Quality OK.
60, Air Quality OK.
60, Air Quality OK.

57, Air Quality OK.
^CTraceback (most recent call last):
```

```
  File "/usr/local/bin/grove_air_quality_sensor_v1_3", line 11, in <module>
    load_entry_point('grove.py==0.5', 'console_scripts', 'grove_air_quality_sensor_v1_3')
()
  File "/usr/local/lib/python3.5/dist-packages/grove/grove_air_quality_sensor_v1_3.py",
line 68, in main
    time.sleep(.1)
KeyboardInterrupt
```

You can use this sensor to detect the air quality. Press ++ctrl+c++ to quit.

NOTE

You may have noticed that for the analog port, the silkscreen pin number is something like **A1, A0**, however in the command we use parameter **0** and **1**, just the same as digital port. So please make sure you plug the module into the correct port, otherwise there may be pin conflicts.

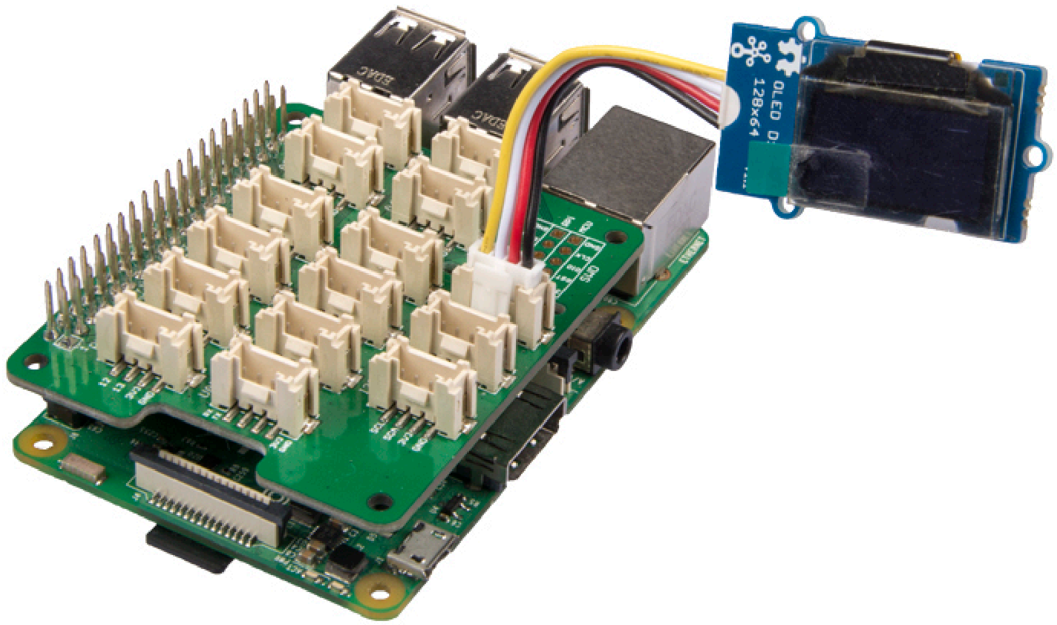
I2C

CAUTION

If you are using **Raspberry Pi with Raspberrypi OS >= Bullseye**, you have to use this command line **only with Python3**.

We will take the Grove - OLED Display 128x64 for example to introduce the I2C port.

Hardware connection diagram



Tap the following command `grove_oled_display_128x64` in the command line interface.

```
(env)pi@raspberrypi:~$ python grove_oled_display_128x64
```

It seems nothing happened, however you can find the most famous sentence in the cyber world if you check your oled. 😄

> **NOTE**
>
> If you use the I2C tool to scan the I2C address of the grove module, you may find two or more address. 0x04 is the adrress of the *Grove Base Hat for Raspberry Pi*.

Schematic Online Viewer

:::noteChange Note:

Because ST32 series chips are out of stock globally, prices have increased several times and there is no clear delivery date. We have no choice but to switch to the MM32 chip. The specific replacement models are as follows: STM32F030F4P6TR is replaced by MM32F031F6P6. After the chip is replaced, the product functions, features, usage methods and codes remain unchanged. It should be noted that the firmware version has changed, and the factory firmware has been adjusted according to different chips. If you need to re-burn the firmware, please download the firmware corresponding to the chip. :::

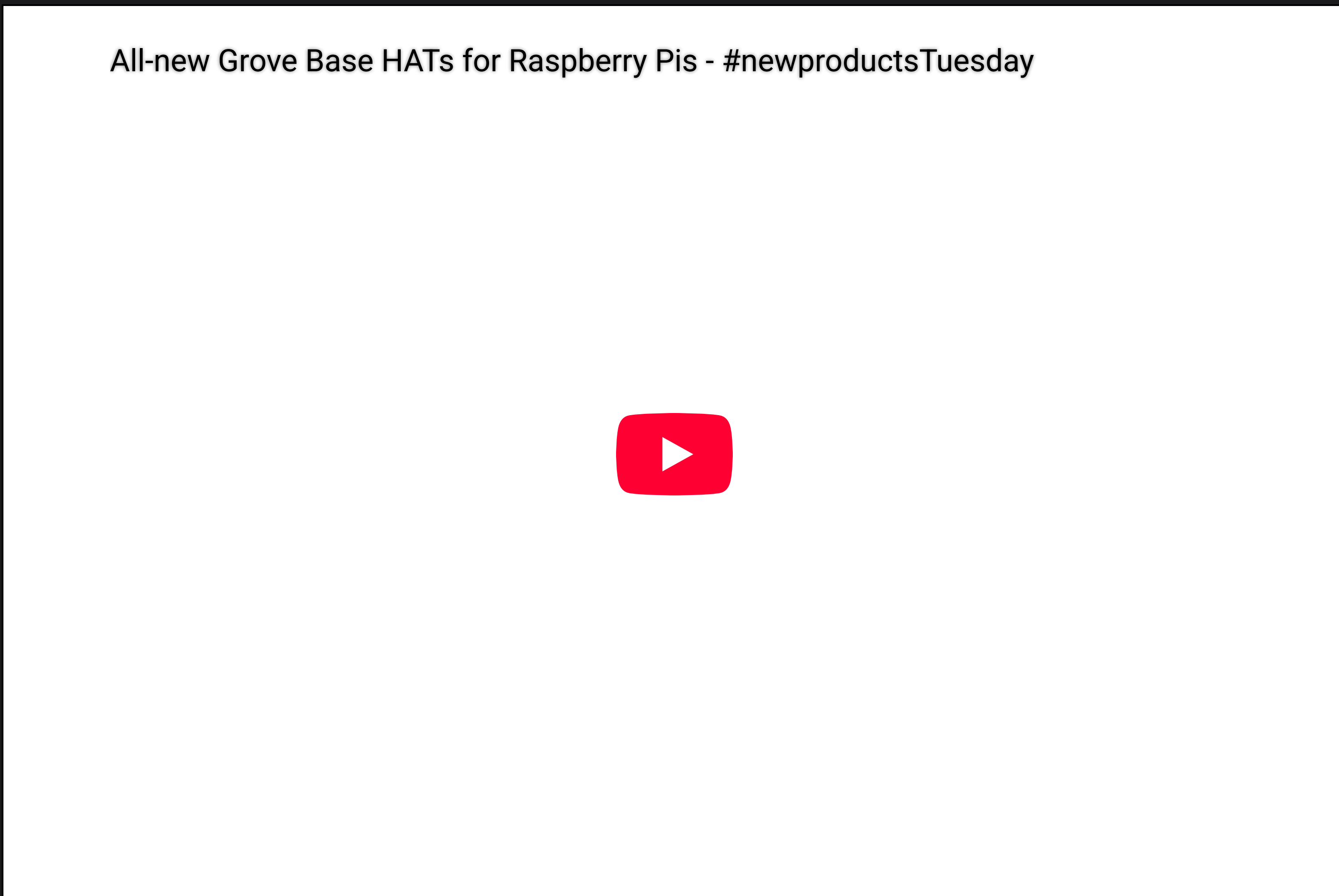
Resources

- **[Zip]** Grove Base Hat for Raspberry Pi Eagle Files
- **[Zip]** Seeed Grove.py Library
- **[Zip]** STM32F030F4P6TR-Firmware
- **[Zip]** MM32F031F6P6-Firmware
- **[PDF]** STM32 Datasheet

- **[PDF]** MM32F031F6P6_Datasheet.pdf

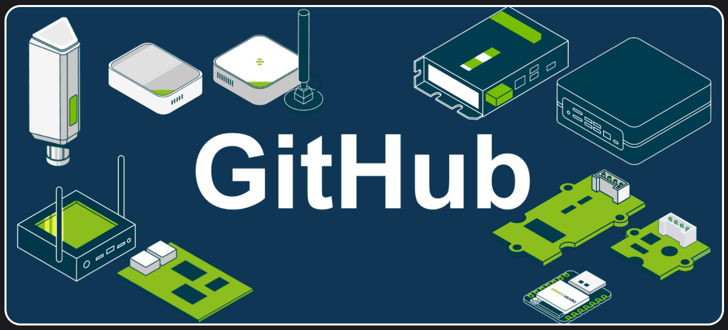
Project

This is the introduction Video of this product.



Tech Support & Product Discussion

Thank you for choosing our products! We are here to provide you with different support to ensure that your experience with our products is as smooth as possible. We offer several communication channels to cater to different preferences and needs.



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Last updated on ***Dec 29, 2025*** *by* ***Brandy***

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